

KUDZANAI CHAKAVARIKA

GIS & Remote Sensing Analyst | Environmental & Climate Data Specialist | Web Mapping Developer

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CORE COMPETENCIES

- Environmental GIS & spatial analysis
- Remote sensing (land cover, vegetation, water)
- Hydrology & dam sedimentation analysis
- Mining impact assessment
- Urban ecology & land use change
- Cartographic design & map layout composition (2D & 3D)
- Web GIS dashboards & data visualization
- Remote sensing image classification

TECHNICAL SKILLS

- QGIS / ArcGIS
- ArcGIS Online
- Google Earth Engine
- Google Earth Pro
- Cesium 3D - 3D map visualization
- Leaflet.js / Web GIS
- Global Mapper
- SEPAL
- Excel & spatial data cleaning

EDUCATION

BSc (Honours) in GIS and Remote Sensing in Natural Resources Management

National University of Science and Technology (NUST), Zimbabwe
2022 - 2026

AREAS OF INTEREST

Climate change adaptation, natural resource management, environmental impact assessment, sustainable urban development and open geospatial data systems.

References available upon request.

PROFESSIONAL SUMMARY

GIS and Remote Sensing graduate with practical, applied experience in environmental monitoring, natural resource analysis, and climate impact assessment across Zimbabwe. Background spans forestry, water resources, mining impact, and urban ecosystems, producing analytical maps, dashboards, and spatial models that support environmental decision-making and sustainable development planning. Skilled in cartographic design, producing clear, well-composed map layouts in both 2D and 3D for technical and public-facing audiences.

WORK EXPERIENCE

GIS intern - Forestry Commission of Zimbabwe

2024 - 2025

- Mapped and analysed 17 gazetted forests across Zimbabwe, assessing spatial distribution and degradation patterns.
- Produced spatial datasets and maps used in environmental conservation reporting and forest monitoring.
- Supported field data collection (GPS) and integration of survey data into GIS systems.

NOTABLE PROJECTS

Chiyadzwa Diamond Fields

Environmental Impact Analysis

Assessed land degradation and ecological stress caused by diamond mining, analysing landscape transformation across the mining zone.

Dam Sedimentation Study

20 Dams, Zimbabwe

Analysed sedimentation trends and water storage loss across more than 20 dams, producing spatial risk maps for water resource management.

Bikita Minerals Mine

Impact on Matezva Dam

Investigated mining influence on downstream water systems, analysing contamination, sediment flow and hydrological risk.

Urban Forest & Wetland Loss

Bulawayo

Mapped land cover changes and wetland degradation across high- and low-density suburbs to quantify urban environmental pressure.

Industrial Gas Emissions Dashboard

Bulawayo

Built an interactive web-based GIS dashboard visualising the spatial distribution of industrial gas emission sources.

Climate Innovation Project

Bloomberg Philanthropies YCAF

Secured and delivered a community-based environmental GIS project in Bulawayo, coordinating spatial analysis and mapping outputs for grant reporting.